

Terminating Differential Amplifiers

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B.1 Introduction

Termination of differential amplifiers is a complex and little understood technique. It has been moved to an appendix because it is a rarely used design problem and so it doesn't slow the flow of material in Chapter 3. It should also be understood that the term *differential amplifier* refers to a single ended op amp configured as a differential amplifier, as opposed to a fully differential op amp configured as a differential amplifier, which is covered separately.

Op amp instrumentation amplifiers have been the mainstay of high accuracy DC instrumentation for decades. The familiar three op amp topology of [Figure B.1](#) has the advantage of high input impedance on both inputs, allowing the designer to interface to sources such as Wheatstone bridges that have relatively high source impedance. The instrumentation amplifier has high gain and is inherently balanced, which also contributes to its popularity.

An instrumentation amplifier is a type of differential amplifier using two input op amps to isolate the source from the input impedance of the differential output stage.

There is a new use for differential amplifiers. They are gaining popularity due to differential data and analog transmission down balanced lines. The traditional instrumentation amplifier and difference amplifier products are too slow to handle these high speed requirements, so new design techniques are required. High speed data can be handled by line receiver products, this appendix focuses on high speed analog.

Balanced line transmission requires termination. This negates the need for high input impedance afforded by the instrumentation amplifier topology of [Figure B.1](#). It allows

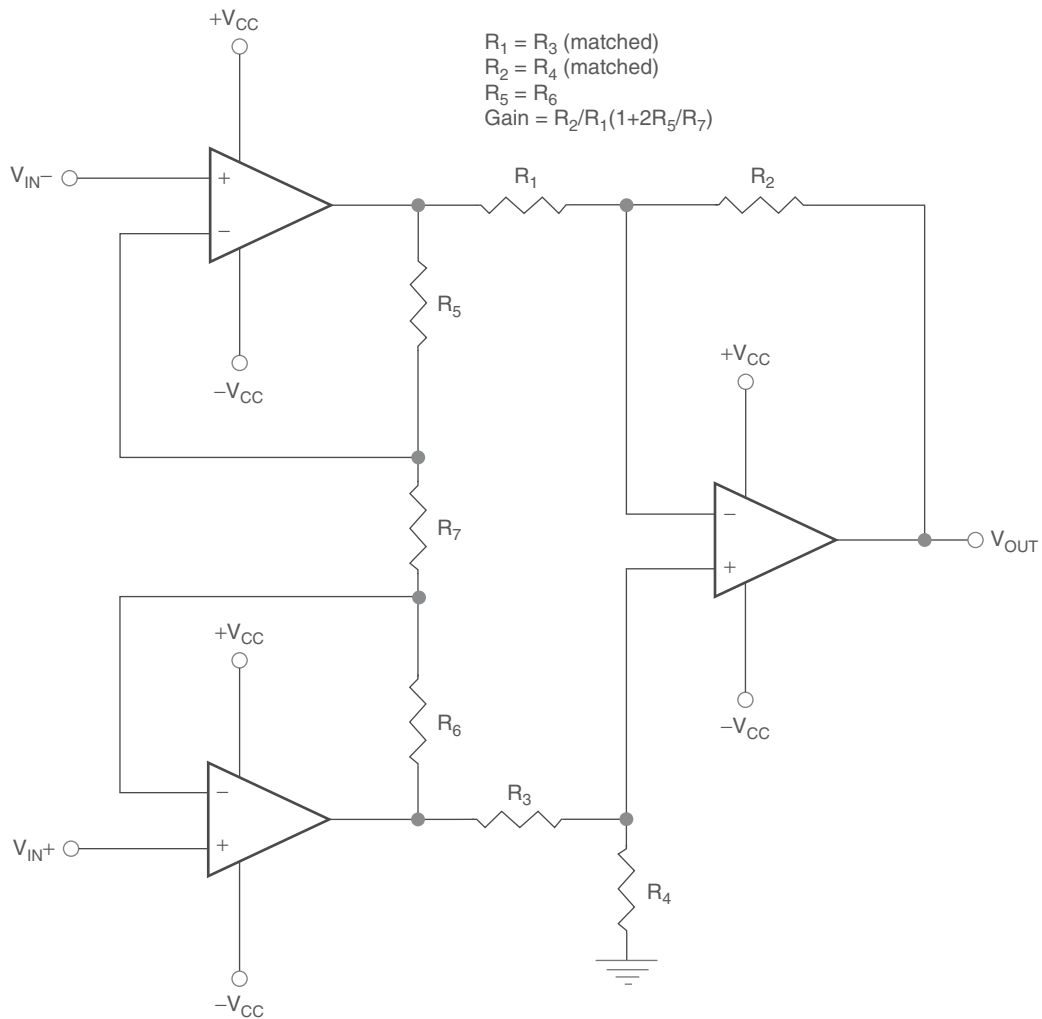


Figure B.1: Instrumentation amplifier.

the designer to delete the two input op amps and implement a differential amplifier consisting of only the third (output) amplifier. There is an additional benefit: The elimination of the two input amplifiers and their associated slew rate limitations allows the one remaining amplifier to receive faster analog signals. The drawback with this system comes with its termination: How does the designer assure proper termination for the balanced line?

B.2 Terminating a Differential Amplifier

To begin with, consider the terminated system shown in [Figure B.2](#).

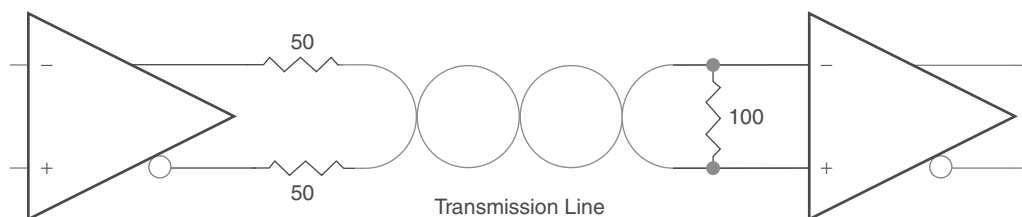


Figure B.2: Data transmission system.

In this system, differential analog data are driven into two $50\ \Omega$ resistors to match a transmission line impedance of $100\ \Omega$. At the far end of the transmission line, a $100\ \Omega$ termination resistor completes the termination, and the terminated input is applied to a receiver.

Next, assume that the balanced line termination can be grounded at its center point; that is, a single fixed resistor that is normally placed from one signal polarity to the other can be tapped in the middle and connected to ground as shown in [Figure B.3](#).

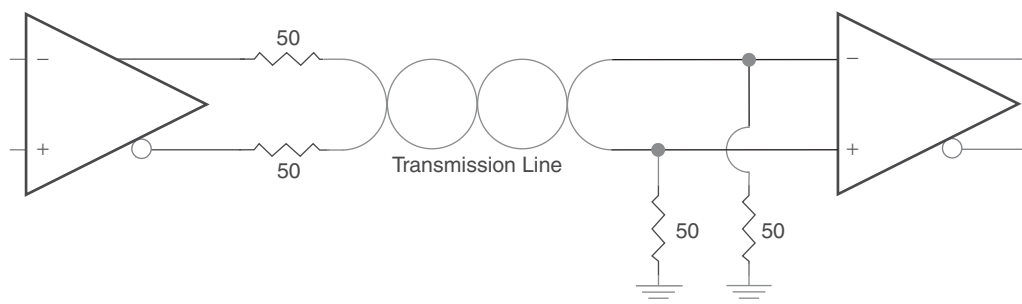


Figure B.3: Modified data transmission system.

Admittedly, this cannot be done in some applications, in particular when the ground at the receiver contains a lot of noise. In practice, however, the line should be

balanced, with equal and opposite voltage swings in each conductor. This means that the midpoint of the signal is at 0 V by definition, no matter what the signal level is. Also, a perfect receiver has infinite common mode rejection. No matter what common mode voltage appears on the inputs (due to the midpoint connection to ground), it is rejected by the receiver. Modern op amps can be chosen that have very good common mode rejection, so this is a good assumption.

Figure B.4 shows the schematic of the proposed high speed receiver. A $100\ \Omega$ line impedance is assumed, because it is a common value of line impedance for cables such as Category 5, 5A, and 6. The input signal is represented by two identical and opposite ideal voltage sources V_1 and V_2 , each in series with a $50\ \Omega$ resistor (for a total of $100\ \Omega$ source impedance).

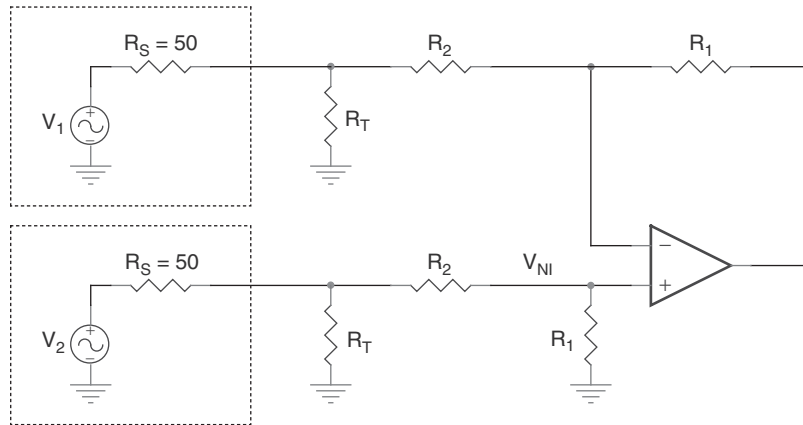


Figure B.4: Differential receiver.

For the receiver to be balanced,

- R_1 top = R_1 bottom.
- R_2 top = R_2 bottom.
- R_T top = R_T bottom.

From superposition, the gains on V_1 and V_2 can be calculated separately, then summed. Therefore, each polarity of the signal is analyzed separately and the results summed.

B.3 Inverting Side

Make $V_2 = 0$.

Gain is the product of a voltage divider and the normal inverting gain equation. The voltage divider is composed of R_S and R_T in parallel with R_2 :

$$\begin{aligned}
 -\frac{V_{\text{OUT}}}{V_1} &= \frac{R_2 \parallel R_T}{R_S + R_2 \parallel R_T} \times \frac{R_1}{R_2} \\
 &= \frac{\frac{R_2 \times R_T}{R_2 + R_T}}{R_S + \frac{R_2 \times R_T}{R_2 + R_T}} \times \frac{R_1}{R_2} \\
 &= \frac{\frac{R_1 \times R_T}{R_2 + R_T}}{R_S + \frac{R_2 \times R_T}{R_2 + R_T}} \\
 &= \frac{\frac{R_1 \times R_T}{R_2 + R_T}}{R_S \times \frac{R_2 + R_T}{R_2 + R_T} + \frac{R_2 \times R_T}{R_2 + R_T}} \\
 &= \frac{R_1 \times R_T}{R_S(R_2 + R_T) + R_2 \times R_T} \\
 \frac{V_{\text{OUT}}}{V_1} &= \frac{-R_1 \times R_T}{R_S \times R_2 + R_S \times R_T + R_2 \times R_T}
 \end{aligned}$$

Therefore,

$$V_{\text{OUT}} = \frac{-V_1 \times R_1 \times R_T}{R_S \times R_2 + R_S \times R_T + R_2 \times R_T}$$

B.4 Noninverting Side

Make $V_1 = 0$.

The noninverting side also has a voltage divider on the input. To simplify matters, first calculate the gain on the voltage at the noninverting input V_{NI} . The gain on this voltage is the normal noninverting gain:

$$\begin{aligned}
 V_{OUT} &= V_{NI} \times \left(1 + \frac{R_1}{R_2 + R_S \parallel R_T} \right) \\
 &= V_{NI} \times \left(1 + \frac{R_1}{R_2 + \frac{R_S \times R_T}{R_S + R_T}} \right) \\
 &= V_{NI} \times \left(1 + \frac{R_1}{R_2 \times \frac{R_S + R_T}{R_S + R_T} + \frac{R_S \times R_T}{R_S + R_T}} \right) \\
 &= V_{NI} \times \left(1 + \frac{R_1 \times (R_S + R_T)}{R_2 \times R_S + R_2 \times R_T + R_S \times R_T} \right) \\
 &= V_{NI} + \frac{V_{NI} \times R_1 \times (R_S + R_T)}{R_2 \times R_S + R_2 \times R_T + R_S \times R_T} \\
 &= \frac{V_{NI} \times (R_2 \times R_S + R_2 \times R_T + R_S \times R_T) + V_{NI} \times R_1 \times (R_S + R_T)}{R_2 \times R_S + R_2 \times R_T + R_S \times R_T} \\
 &= \frac{V_{NI} \times R_2 \times R_S + V_{NI} \times R_2 \times R_T + V_{NI} \times R_S \times R_T + V_{NI} \times R_1 \times R_S + V_{NI} \times R_1 \times R_T}{R_2 \times R_S + R_2 \times R_T + R_S \times R_T}
 \end{aligned}$$

Therefore,

$$V_{OUT} = V_{NI} \times \frac{R_2 \times R_S + R_2 \times R_T + R_S \times R_T + R_1 \times R_S + R_1 \times R_T}{R_2 \times R_S + R_2 \times R_T + R_S \times R_T}$$

Next, calculate V_{NI} from the voltage divider rule. There are two voltage dividers on V_2 : one composed of R_S and the parallel combination of $R_T \parallel (R_1 + R_2)$, and the second composed of R_1 and R_2 .

First, define $R_P = R_T \parallel (R_1 + R_2)$:

$$V_{NI} = V_2 \times \left(\frac{R_P}{R_S + R_P} \right) \times \left(\frac{R_1}{R_1 + R_2} \right)$$

Substituting for R_P ,

$$\begin{aligned} V_{NI} &= V_2 \times \left(\frac{\frac{R_T \times (R_1 + R_2)}{R_T + R_1 + R_2}}{R_S + \left(\frac{R_T \times (R_1 + R_2)}{R_T + R_1 + R_2} \right)} \right) \times \frac{R_1}{R_1 + R_2} \\ &= V_2 \times \left[\frac{R_T \times R_1}{R_S \times (R_T + R_1 + R_2) + R_T \times (R_1 + R_2)} \right] \end{aligned}$$

Therefore,

$$V_{NI} = \frac{V_2 \times R_1 \times R_T}{R_S \times R_T + R_S \times R_1 + R_S \times R_2 + R_T \times R_1 + R_T \times R_2}$$

Now, substitute the expression for V_{NI} into the expression for the noninverting gain:

$$\begin{aligned} V_{OUT} &= \frac{V_2 \times R_1 \times R_T}{R_S \times R_T + R_S \times R_1 + R_S \times R_2 + R_T \times R_1 + R_T \times R_2} \\ &\times \frac{R_2 \times R_S + R_2 \times R_T + R_S \times R_T + R_1 \times R_S + R_1 \times R_T}{R_2 \times R_S + R_2 \times R_T + R_S \times R_T} \end{aligned}$$

and the large numerator and denominator cancel out; therefore,

$$V_{OUT} = \frac{V_2 \times R_1 \times R_T}{R_2 \times R_S + R_2 \times R_T + R_S \times R_T}$$

B.5 Differential Output

To get the differential output voltage, just add together the inverting and noninverting gains using the superposition rule:

$$\begin{aligned} V_{\text{OUT}} &= \frac{V_2 \times R_1 \times R_T}{R_2 \times R_S + R_2 \times R_T + R_S \times R_T} - \frac{V_1 \times R_1 \times R_T}{R_S \times R_2 + R_S \times R_T + R_2 \times R_T} \\ &= \frac{(V_2 - V_1) \times R_1 \times R_T}{R_2 \times R_S + R_2 \times R_T + R_S \times R_T} \end{aligned}$$

Therefore,

$$\frac{V_{\text{OUT}}}{V_2 - V_1} = \frac{R_1 \times R_T}{R_S \times R_2 + R_S \times R_T + R_2 \times R_T}$$

This equation is a bit formidable looking. Fortunately, three out of the five variables are known:

- R_S is the source resistance of the line (usually half of the line impedance, since it is tapped in the middle; so for a 100 Ω line, R_S is 50 Ω on each side, for a total of 100 Ω).
- The gain is known: $V_{\text{OUT}}/(V_2 - V_1)$.
- The value of R_1 is known, because the data sheet usually specifies a value of feedback resistance for maximum stability.
- This leaves only two unknowns: the input resistance R_2 and proper termination resistance R_T . The calculation, therefore, becomes a goal seeking problem in two variables. A Javascript design utility has been located on the Texas Instruments Web site (ti.com) for the convenience of the designer.

B.6 Testing the Result

The preceding expressions were tested at a gain of 0.5 and 1.

B.6.1 Gain of 0.5

$$\begin{aligned}\frac{V_{OUT}}{V_2 - V_1} &= \frac{R_1 \times R_T}{R_S \times R_2 + R_S \times R_T + R_2 \times R_T} \\ &= \frac{249 \times 61.9}{50 \times 249 + 50 \times 61.9 + 249 \times 61.9} \\ &= \frac{15,413.1}{12,450 + 3,095 + 15,413.1} = \frac{15,413.1}{30,958.1} = 0.498 = -6.058 \text{ dB}\end{aligned}$$

For inverting input with noninverting input grounded, simulation and lab results are shown in Figure B.5.

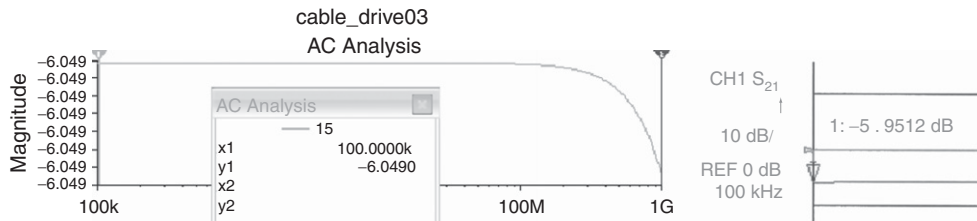


Figure B.5: Simulation and lab results for inverting input, gain of 0.5.

For noninverting input with inverting input grounded, simulation and lab results are shown in Figure B.6.

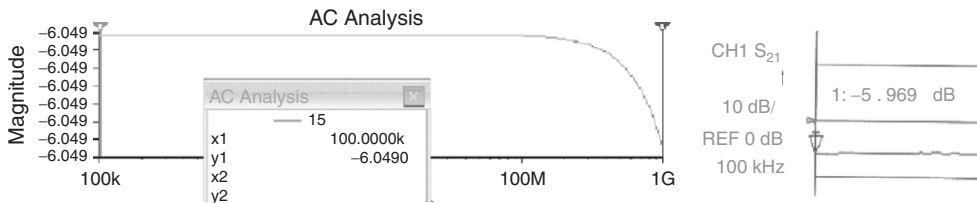


Figure B.6: Simulation and lab results for noninverting input, gain of 0.5.

B.6.2 Gain of 1

Next, at a stage gain of 1,

$$\begin{aligned}\frac{V_{OUT}}{V_2 - V_1} &= \frac{R_1 \times R_T}{R_S \times R_2 + R_S \times R_T + R_2 \times R_T} \\ &= \frac{499 \times 61.9}{50 \times 249 + 50 \times 61.9 + 249 \times 61.9} \\ &= \frac{30,888.1}{12,450 + 3,095 + 15,413.1} = \frac{30,888.1}{30,958.1} = 0.998 = -0.02 \text{ dB}\end{aligned}$$

For inverting input with noninverting input grounded, simulation and lab results are shown in [Figure B.7](#).

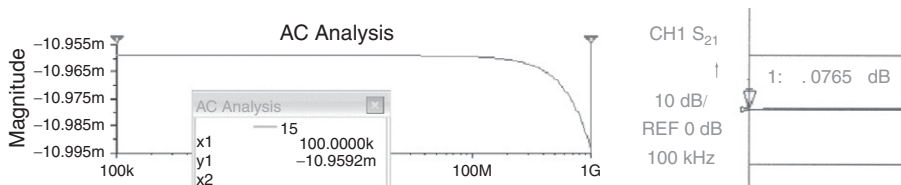


Figure B.7: Simulation and lab results for inverting input, gain of 1.

For noninverting input with inverting input grounded, simulation and lab results are shown in [Figure B.8](#).

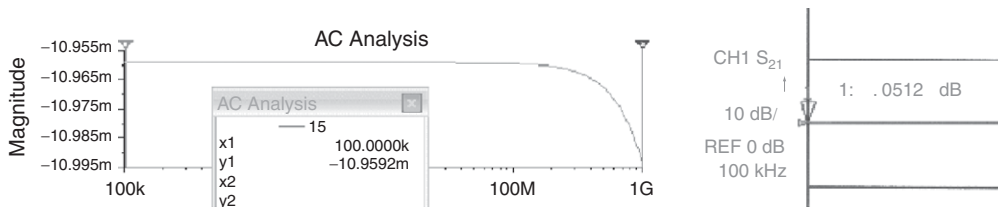


Figure B.8: Simulation and lab results for noninverting input, gain of 1.

Again, pretty close. At this point, it is important to point out that, when the inputs are equal in magnitude and opposite in polarity, the stage gain is double the results seen here. In other words, stage gain on a differential input is double the stage gain when only one input is used.